

Executive Summary for ShopLC's Executives

ShopLC's return rates vary widely across product families, even after accounting for price, discounting, and TV/online exposure. Our analysis recommends focusing first on the codenames whose return rates are most misaligned with expectations. Both the families with unusually high returns and those with unusually low returns. Addressing these two sets of products offers the most straightforward path to reducing returns without limiting sales. Coneflower, Peony, Zinnia, Ranunculus, and Tulip return significantly more often than expected based on their characteristics. In contrast, Daffodil, Lilac, Canna Lily, Nigella, and Hellebore return far less often than expected. These groups should drive immediate quality reviews, presentation adjustments, and policy experiments tailored by category.

1. Product Families with Higher than Expected Return Rates

The following codenames have return rates well above what the model predicts, even after accounting for price, discounting, and exposure:

Codename	Actual Avg Return Rate (%)	Expected (%)	Difference	Weekly Records
Coneflower	28.1	24.5	+3.6	605
Peony	20.5	17.6	+2.9	586
Zinnia	25.1	22.7	+2.4	964
Ranunculus	17.5	15.3	+2.2	824
Tulip	21.5	19.3	+2.2	519

These consistent gaps indicate consistent, predictable issues rather than isolated spikes in returns. Recommended actions include:

- **Quality & durability checks:** Compare the failure or wear patterns of these items with those of low-return products.
- **On-air and online presentation review:** Confirm that visual appearance, color, sizing, and accessory expectations are accurately conveyed.
- **Tighter expectation setting in descriptions:** Clarify fit, materials, included components, or assembly needs where customers may be surprised upon arrival.

2. Product Families with Lower Than Expected Return Rates

These codenames perform far better than the model's predictions and should be studied for consistently performing better:

Codename	Actual Avg Return Rate (%)	Expected (%)	Difference	Weekly Records
Daffodil	4.4	10.9	-6.5	2,854

Lilac	7.2	10.9	-3.7	752
Canna Lily	5.7	9.2	-3.5	811
Nigella	7.7	11.1	-3.4	879
Hellebore	27.8	30.4	-2.6	593

Recommended actions include:

- **Identify what these families do well:** Review their photography, descriptions, packaging, and on-air demos to determine what consistently helps customers make accurate purchase decisions.
- **Create presentation templates based on these codenames:** Use their successful approaches (image angles, fit notes, host talking points, etc.) as the standard for related categories—especially high-return families.
- **Prioritize these products for high-visibility promotions:** Their reliably low return rates make them strong candidates for deep discounts, Today’s Top Deals, and longer TV segments.
- **Exclude these families from early return-policy experiments:** Keep high-performing families stable while testing changes on higher-risk products.

3. Recommendations for Return-Policy Adjustment

- Require return-reason selection at initiation. Customers returning high-risk families (e.g., Coneflower, Peony, Zinnia, Ranunculus, Tulip) must choose from standardized reasons such as: “Damaged,” “Not as shown,” “Fit/size issue,” “Missing parts,” “Changed mind,” “Quality concern.”
- Use return-reason data to refine product pages and on-air presentations.
- Introduce a small processing fee only for “changed mind” returns in high-risk categories. A \$2–\$4 processing fee (excluding damage, defects, or presentation issues) can reduce impulse buying in families that already exceed expected return rates by 2 to 4 percentage points.

This preserves full refunds for genuine problems while discouraging serial “try-and-return” behavior in costly product lines.

4. Additional Data Needed for Better Return-Rate Insights

To understand why certain products are returned more or less often than expected, ShopLC should begin capturing a few additional fields during the purchase and return processes. These data points would help distinguish quality issues from expectation gaps or shipping problems:

- Reason for return (quality issue, damaged, not as shown, wrong size/color, changed mind).
- Customer-selected variation (size, color, style) to flag consistent mismatch patterns.
- Arrival/packaging condition photos to identify damage or missing components.

Collecting these fields would enable ShopLC to pinpoint the key drivers behind high-return families and replicate what works in low-return families, resulting in more accurate and actionable improvements.

Key Drivers of Return Rate

Return rates across the 40 product families range from about 4% to 35%, and several consistent, measurable drivers influence these differences.

1. Exposure (showings, TV duration, viewership)

Products with more on-air exposure tend to be returned more often.

- Codenames with minimal exposure typically yield returns of 14–18%.
- Codenames with heavy exposure exhibit return rates closer to 20–25%, representing a 3–6 percentage point increase.

This pattern reflects the effect of marginal buyers. Wider reach attracts shoppers who are less certain, thereby increasing returns.

2. Unit Cost

Higher unit-cost items show noticeably higher return rates.

- Lowest-cost items average around 12–15%.
- Mid-range items fall in the 17–22% range.
- The highest-cost items reach 25–30%, roughly 10–15 percentage points higher than the lowest tier.

Customers hold expensive items to higher standards and return them more readily if expectations are not met.

3. Discounting and Shipping Promotions

More aggressive discounts correspond to higher return rates.

- Low-discount or no-discount weeks average around 15–18% returns.
- Heavy-discount weeks reach 18–22%, representing a 2–4 point increase.

This supports the idea of “trial-based buying,” where customers order more freely when prices drop but return items that don’t fully satisfy them.

Exposure, product cost, and discount depth each shift returns by several percentage points, showing that not just what ShopLC sells, but how it is sold, plays a significant role in return behavior.

Date Summary

This analysis is based on 617,904 weekly records spanning from June 2021 to August 2025. Each row represents a product family within a specific price range for a particular week. The dataset includes information about price, cost of goods, markup, product discounts, shipping discounts, variety count, number of showings, TV duration, estimated viewership, and both gross and net quantities sold, from which return rates are calculated. Records with impossible return rates (below 0% or above 100%) were removed to maintain a realistic evaluation.

Limitations of This Analysis

- Weekly aggregation may mask short-term spikes (e.g., post-promotion surges) and occasionally inflate or deflate the return rate.

- Missing return-reason data hinders root-cause diagnosis. Without details like “damaged,” “fit/size issue,” etc., it’s hard to identify whether high return rates stem from quality issues, expectation gaps, or shipping problems.
- No customer-level variation fields (such as size, color, or style) are included in the dataset, which prevents us from detecting mismatch patterns that may explain deviations for certain product families.

Technical Report

Model Used

To estimate the expected return rate for each Codename–PriceBin–Week combination, a Random Forest regression model was trained on the cleaned training data. The model included the following predictors:

- Codename (categorical; 40 product families)
- PriceBin
- Discount variables (product discount, shipping discount)
- Exposure variables (TV showings, total airtime seconds, estimated viewership)
- Cost and pricing variables (UnitCost, CostGoods, Markup)
- Variety count
- Date variables (week, month, year)

The Random Forest was used with standard parameters (default number of trees, mtry optimized internally). The goal was not forecasting, but establishing a baseline expected return rate so that unusually high or low products could be identified.

Model Performance on Holdout Data

Model accuracy was evaluated on a holdout sample not used in training.

- Naïve baseline RMSE (predicting the overall average return rate for all rows):
 - 0.1040
- Random Forest RMSE:
 - 0.0677

This represents a 35% reduction in prediction error, indicating that the Random Forest provides a significantly more accurate estimate of expected return rates.

- Holdout R^2 : 0.577
- (The model explains over half of the variation in weekly return rates.)

This performance is strong enough to reliably distinguish codenames whose return behavior differs substantially from what the model expects.

Comparison Dataframe (Actual vs. Expected Return Rates)

The table below contains all 40 codenames:

	Codename	ActualAvg	PredAvg	n	Deviation
	<fct>	<dbl>	<dbl>	<int>	<dbl>
1	Allium	10.4	10.6	734	-0.200
2	Anthurium	21.6	21.7	830	-0.1000
3	BlackEyedSusan	21.3	22.6	914	-1.3
4	Bleedingheart	15.2	14.2	1156	1
5	Buttercup	20.1	19.6	1207	0.5
6	Camellia	19.6	19.4	893	0.200
7	CannaLily	5.6	9.2	794	-3.6
8	Chrysanthemum	18.6	19.2	694	-0.600
9	Coneflower	28.2	24.3	596	3.9
10	Cosmos	19.8	20.2	2521	-0.400
11	Daffodil	4.3	10.8	2830	-6.5
12	Daisy	21.1	18.8	693	2.3
13	Dianthus	19.2	19.1	568	0.1000
14	DustyMiller	21	19.7	768	1.3
15	DutchIris	19.9	20.6	1280	-0.700
16	ForgetMeNot	20.9	20	1493	0.900
17	Freesia	22.4	21.8	708	0.600
18	Fuchsia	19.7	19.2	800	0.5
19	Gaillardia	16.3	16.3	1687	0
20	Geranium	34.7	33.6	1044	1.10
21	Gladiolus	25.5	23.7	2558	1.8
22	Hellebore	27.5	30.2	606	-2.7
23	Hyacinth	33.2	32.9	657	0.300
24	Hydrangea	19.4	18.5	2148	0.900
25	Jasmine	26.2	28	678	-1.8
26	Lilac	7.1	11.1	813	-4
27	Lily	20.5	19.8	1138	0.700
28	Lupine	17.8	17.4	1784	0.400
29	Nigella	8	11	808	-3
30	Orchid	25	22.8	1324	2.2
31	Passionflower	19.3	16.9	517	2.40
32	Peony	21	17.4	571	3.6
33	Philadelphus	17.1	16.9	1469	0.200
34	Protea	16.9	18.3	952	-1.40
35	Ranunculus	17.3	15	831	2.3
36	Stephanotis	20.8	20.5	635	0.300
37	Sweetpea	13.9	13.2	591	0.700
38	Tulip	21.1	18.9	565	2.20
39	Wallflower	19.1	18.1	671	1
40	Zinnia	25.1	22.8	942	2.3

- Codename, column 1
- Actual average return rate (%), column 2

- Model-predicted (expected) return rate (%), column 3
- Number of weekly records (n), column 4
- Deviation (Actual – Expected), column 5

Red values indicate lower-than-expected return rates, while black values indicate higher-than-expected return rates.

Interpretation of the Comparison Table

High-Deviation Families (Actual > Expected)

Families exceeding expected returns by 2+ percentage points include:

- Coneflower (+3.9)
- Peony (+3.6)
- Passionflower (+2.4)
- Zinnia (+2.3)
- Ranunculus (+2.3)
- Tulip (+2.2)
- Orchid (+2.2)
- Daisy (+2.3)

These represent the clearest targets for quality checks and presentation adjustments.

Low-Deviation Families (Actual < Expected)

Families performing significantly better than expected include:

- Daffodil (−6.5)
- Lilac (−4.0)
- CannaLily (−3.6)
- Nigella (−3.0)
- Hellebore (−2.7)

These are reliable, low-return categories suitable for promotions, modeling best practices, and policy stability.

Limitations of the Analysis

Although the model performs strongly, several limitations apply:

- The analysis uses only the fields available in the dataset. Missing variables, such as customer reasoning, product condition upon arrival, and variation-specific issues (e.g., size, color), limit diagnostic accuracy.
- The data reflect product families, not individual items. Differences within a family may be hidden.
- Behavioral factors are not included. Customer loyalty, shopping history, and purchasing context (seasonal vs. urgent buying) are unknown.
- Return rates are calculated weekly. Timing mismatches (delayed returns) may introduce small amounts of noise.

These limitations do not undermine the broad conclusions but suggest where future data collection efforts would yield better insights